CUBIC SUPERCONGRUENCES FOR A DIXON-TYPE SUM, FRACTIONAL FACTORIAL RATIOS AND TWO APÉRY-LIKE SEQUENCES

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Abstract. We prove several supercongruences of the form $a(np^r) \equiv a(np^{r-1}) \pmod{p^{3r}}$, $p \geq 5$, conjectured by P. Bala on the OEIS. For the Dixon-type sum $a(n) = \sum_k (-1)^k \binom{n}{k}^2 \binom{n-2}{k}$ (A361711) we prove the tower for every odd n, from an exact product identity over the units below $np^k/2$, and show that it fails for even n. We prove that the fractional factorial ratios A364172, A364178 and A364179 are integers, which was conjectural, and, through a unit-product principle that reduces the towers for A364172–A364184 to integrality, that these and A364175 satisfy the tower. For the Apéry-like sequence A363983 we prove the tower by the method of Osburn, Sahu and Straub. For the two-index sum A376462 we prove both the tower and the shifted congruence $A(np^r - 1) \equiv A(np^{r-1} - 1) \pmod{p^{3r}}$; the latter extends Beukers' 1985 block induction with carry labels, carry-restricted harmonic sums and moment kernels. Along the way we repair two steps of Beukers' printed argument. The main theorems are formalised in Lean 4 with Mathlib.

1. Introduction

For a sequence of integers $a(n)$ and a prime p, congruences of the form

$$a(np^r) \equiv a(np^{r-1}) \pmod{p^{3r}} \qquad (n, r \geq 1) \tag{1}$$

are called *(cubic) supercongruences*. The model case is $a(n) = \binom{2n}{n}$, for which (1) holds for $p \geq 5$ by the theorem of Jacobsthal and Kazandzidis [6, 13, 22]; for $n = r = 1$ it is Wolstenholme's $\binom{2p}{p} \equiv 2 \pmod{p^3}$ [25]. Many such congruences were recorded as conjectures on the OEIS [17] by P. Bala [2]. This paper settles several of them and a related integrality question.

A Dixon-type sum. Let $a(1) = 1$ and

$$a(n) = \sum_{k=0}^{n-2} (-1)^k \binom{n}{k}^2 \binom{n-2}{k} \qquad (n \geq 2),$$

so $(a(n))_{n\geq 1} = 1, 1, -8, 5, 126, -168, -2400, 4125, \ldots$ (A361711). Bala conjectured $a(p^k) \equiv a(p^{k-1}) \pmod{p^{3k}}$ for $p \geq 5$.

Theorem 1.1. *Let $p \geq 5$ be prime and $k \geq 1$.*

(1) For every odd $n \geq 1$: $a(np^k) \equiv a(np^{k-1}) \pmod{p^{3k}}$. More precisely, if $p \nmid n$, $M = np^k$ and $U = \{u \in \mathbb{Z} : 0 < u < M/2,\ p \nmid u\}$, then

$$\frac{a(np^k)}{a(np^{k-1})} = \prod_{u \in U} \Big(1 - \frac{M^2}{u^2}\Big). \tag{2}$$

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(2) The congruence in (1) is false in general for even n: $a(10) - a(2) = -98099 \not\equiv 0 \pmod 5$.

The case $n = 1$ is Bala's conjecture. The exponent $3k$ is optimal: $v_p(a(p^k) - a(p^{k-1})) = 2k + v_p(H)$ with $H = \sum_{u \in U} u^{-2}$, and in all computed cases $v_p(H) = k$, except $p = 7$, where $v_7(H) = k + 1$ (Section 10).

Fractional factorial ratios. With $x! := \Gamma(1 + x)$ consider

$$\mathrm{A364175}(n) = \frac{(6n)!\,(2n/3)!}{(3n)!\,(2n)!\,(5n/3)!}, \qquad \mathrm{A364178}(n) = \frac{(10n)!\,(3n)!\,(n/2)!}{(6n)!\,(5n)!\,(3n/2)!\,n!},$$

$$\mathrm{A364172}(n) = \frac{(6n)!\,(n/3)!}{(3n)!\,(2n)!\,(4n/3)!}, \qquad \mathrm{A364179}(n) = \frac{(10n)!\,(n/2)!}{(5n)!\,(4n)!\,(3n/2)!}.$$

On the OEIS these are listed as "only conjecturally" integer sequences; Radcliffe [20] proved that A364175 is integral. For even $n = 2m$, A364178($2m$) is the Vasyunin ratio

$$\frac{(20m)!\,(6m)!\,m!}{(12m)!\,(10m)!\,(3m)!\,(2m)!}$$

studied by Bober [5] (A295470).

Theorem 1.2. *A364178, A364172 and A364179 are integer sequences.*

Theorem 1.3. *For every prime $p \geq 5$ and all $n, r \geq 1$, each of A364175, A364178, A364172 and A364179 satisfies* (1).

Theorem 1.3 is a special case of a ratio tower for a whole family of fractional factorial ratios (Theorem 5.2), which reduces Bala's conjectures on the thirteen entries A364172–A364184 to integrality.

Two Apéry-like sequences. The Apéry numbers $\sum_k \binom{n}{k}^2\binom{n+k}{k}$ satisfy, for $p \geq 5$, both

$$a(mp^r) \equiv a(mp^{r-1}) \quad \text{and} \quad a(mp^r - 1) \equiv a(mp^{r-1} - 1) \pmod{p^{3r}}; \tag{3}$$

the second, *shifted* form is due to Beukers [3] and the first to Coster [8] (see also Gessel [11]). Osburn, Sahu and Straub [18] proved the first congruence for several sporadic sequences, and Straub [23] explained both forms for a multivariate family through negative indices. P. Bala [2] conjectured the first congruence for A363983 and the pair (3) for A376462.

Theorem 1.4. *Let $a(n) = \sum_{k=\lfloor (n+1)/2 \rfloor}^{n} (-1)^{n+k}\binom{n}{k}\binom{n+k-1}{k}\binom{2k}{n}$ (A363983). For every prime $p \geq 5$ and all $m, r \geq 1$, $a(mp^r) \equiv a(mp^{r-1}) \pmod{p^{3r}}$.*

Theorem 1.5. *Let $A(n) = \sum_{k=0}^{n} \binom{n}{k}^2\binom{n+k}{k} T(n, n-k)$ with $T(n, j) = \sum_i \binom{n}{i}^2\binom{n+j-i}{j-i}$ (A376462, A108625). For every prime $p \geq 5$ and all $m, r \geq 1$,*

$$A(mp^r) \equiv A(mp^{r-1}) \pmod{p^{3r}} \qquad \text{and} \qquad A(mp^r - 1) \equiv A(mp^{r-1} - 1) \pmod{p^{3r}}.$$

A363983 is not an instance of [18, Thm. 1.2]: its summand contains $\binom{n}{k}$ only once, a factor $\binom{n+k-1}{k}$ instead of $\binom{n+k}{k}^B$, and a sign. The point is that $\binom{n+k-1}{k} = \frac{n}{k}\binom{n+k-1}{k-1}$ supplies the second power of p that OSS obtain from $\binom{n}{k}^2$. The signed analogue $\sum_k (-1)^{n+k}\binom{n}{k}^2\binom{2k}{n}$ *fails* the tower ($v_5(f(5) - f(1)) = 2$), so the choice of weights matters.

$A(n)$ is the constant term of Λ^n for the Laurent polynomial

$$\Lambda = \frac{(1 + u + v)(1 + z)(1 + u + z)(1 + w)(1 + v + w)}{uvzw},$$

whose Newton polytope has a unique interior lattice point. General theory then gives the Gauss congruences and Dwork-type congruences [15, 4], but not the exponent $3r$. No identity of the type $A(-n) = A(n-1)$ holds, and A376462 is not in Straub's family, so the shifted congruence needs a direct proof. We give one by extending Beukers' method (Section 8). The unshifted congruences follow the template of [18] (Sections 6 and 7).

Beukers' printed argument. Our proof re-proves two lemmas of [3]. Checked against the published text, the printed proof of his Lemma 4(i) reduces it to an auxiliary congruence that is false, and his Lemma 4(ii) is stated modulo p^{t+s}, which is too weak for its use on p. 154 (Section 9). His theorem is unaffected; the repairs are Lemmas 8.4 and 8.5.

Method. All proofs follow one pattern. The indices of a sum, or the factors of a product, are split into multiples of p and units. The part indexed by multiples of p is compared with the same sequence at the lower level np^{r-1}, and the unit part is controlled by Wolstenholme-type congruences for sums over units (Section 2). The comparison is an exact product identity for A361711 (Lemma 3.2), a unit-product principle for the factorial ratios (Section 5), and the digit-dropping identities of Lemma 2.6 for the Apéry-like sequences. For the shifted congruence of A376462 the unit part no longer carries enough powers of p by itself, and the block inductions of Section 8 take its place.

Earlier claims. Unreviewed drafts by rbajaj5 [21] claim the tower of A364175 given Radcliffe's integrality (1 August 2026), the reduction of the towers of A364172–A364184 to integrality by a rational gamma-ratio transfer (same date, our Theorem 5.2), and the integrality and towers of A364178 and A364179 as half-index cases of A295470 and A295471 (13 August 2026). Our proofs were written independently of these drafts, are complete, and for A364175 and A364178 are machine-checked. Theorem 1.1, the integrality of A364172, and the failure for even n are not claimed there.

Machine verification. Theorem 1.1 for $n = 1$, Theorem 1.3 for A364175 and A364178 (including the integrality of A364178), and Theorems 1.4 and 1.5 are formalised in Lean 4 [9] with Mathlib [24]; the formal statements are those of the Formal Conjectures benchmark, verbatim, and the proofs use only the standard axioms (Section 10).

Notation. p is a prime, v_p the p-adic valuation on $\mathbb{Q}$, and congruences between rationals with denominators prime to p are understood in $\mathbb{Z}_{(p)}$. A prime on a sum or product ($\sum'$, $\prod'$) restricts the index to integers prime to p. Binomials are generalised: $\binom{a}{b} = a(a-1)\cdots(a-b+1)/b!$ for $a \in \mathbb{Z}$, $b \geq 0$, and $\binom{a}{b} = 0$ for $b < 0$. Two identities are used throughout: $\binom{a}{b} = \frac{a}{b}\binom{a-1}{b-1}$ ($b \geq 1$) and $\binom{n+b-1}{b} = (-1)^b\binom{-n}{b}$. For $t, l \geq 0$ the *block* is $B_t(l) = \{k \geq 0 : \lfloor k/p^t \rfloor = l\}$. $H'_n = \sum'_{\lambda \leq n} 1/\lambda$, $\sigma_t(n) = \sum'_{k \in B_t(n)} 1/k$ and $S_d(x) = \sum'_{\mu \leq x} \mu^{-d}$. For $e \geq 1$ let $R_e = \{u : 0 < u < p^e,\ p \nmid u\}$; we write S_j for $S_j(p^e) = \sum_{u \in R_e} u^{-j}$ when e is clear from the context. From Section 6 on, $p \geq 5$.

2. Sums over units and binomial tools

Lemma 2.1. *Let $p \geq 5$ and $e \geq 1$. Then $S_2 \equiv 0 \pmod{p^e}$ and $S_1 \equiv 0 \pmod{p^{2e}}$. Moreover, for every integer $N = np^e$ with $n \geq 1$,*

$$\sum_{0<k\leq N}{}' \frac{1}{k} \equiv 0 \pmod{p^{2e}}, \qquad \sum_{0<k\leq N}{}' \frac{1}{k^2} \equiv 0 \pmod{p^e}.$$

Proof. Since $u \mapsto u^{-1} \bmod p^e$ permutes R_e,

$$S_2 \equiv \sum_{u \in R_e} u^2 = \frac{(p^e-1)p^e(2p^e-1)}{6} - p^2\,\frac{(p^{e-1}-1)p^{e-1}(2p^{e-1}-1)}{6} \equiv 0 \pmod{p^e},$$

as 6 is a unit. Pairing u with $p^e - u$ gives $\frac{1}{u} + \frac{1}{p^e-u} = \frac{p^e}{u(p^e-u)} \equiv -\frac{p^e}{u^2} \pmod{p^{2e}}$, hence $2S_1 \equiv -p^e S_2 \equiv 0 \pmod{p^{2e}}$. For the sums up to N write $k = \beta p^e + u$ with $0 \le \beta < n$, $u \in R_e$, and use $\frac{1}{\beta p^e+u} \equiv \frac{1}{u} - \frac{\beta p^e}{u^2} \pmod{p^{2e}}$ and $\frac{1}{(\beta p^e+u)^2} \equiv \frac{1}{u^2} \pmod{p^e}$. □

Lemma 2.2. *Let $p \ge 5$, $k \ge 1$, n odd with $p \nmid n$, $M = np^k$ and U as in Theorem 1.1. Then $H := \sum_{u \in U} u^{-2} \equiv 0 \pmod{p^k}$.*

Proof. M is odd, so $u \mapsto M-u$ maps U bijectively onto the units in $(M/2, M)$, and $(M-u)^{-2} \equiv u^{-2} \pmod{p^k}$. Hence $2H \equiv \sum'_{0<v<M} v^{-2} \equiv 0 \pmod{p^k}$ by Lemma 2.1. □

Lemmas 2.4 and 2.5 below extend Lemma 2.1 to sums over blocks; Sections 3–5 need only Lemma 2.1.

Lemma 2.3 (Kazandzidis; Robert–Zuber [22])**.** *For $p \ge 5$ and integers $A > B > 0$: $\binom{pA}{pB}/\binom{A}{B} \in 1 + p^3AB(A-B)\mathbb{Z}_p$.*

Consequently [18, Lemma 2.1], if $p \nmid ab$ and $r, s \ge 1$, then $\binom{p^r a}{p^s b}/\binom{p^{r-1}a}{p^{s-1}b} \equiv 1 \pmod{p^{r+s+\min(r,s)}}$ whenever the binomials are nonzero.

Lemma 2.4 (power sums over units)**.** *Let $v, d \ge 1$ and $e(d) = 0$ if $(p-1) \nmid d$, $e(d) = 1+v_p(d)$ otherwise. Then $\sum_{u \in (\mathbb{Z}/p^v)^\times} u^{-d} \equiv 0 \pmod{p^{v-e(d)}}$. In particular $S_2(x) \equiv 0 \pmod{p^v}$ whenever $p^v \mid x$, and the sum of u^{-2} over the units of any block $B_v(l)$, or of either half of it, is $\equiv 0 \pmod{p^v}$.*

Proof. Let g generate $(\mathbb{Z}/p^v)^\times$. Then $(g^{-d}-1)\sum u^{-d} \equiv 0$. If $(p-1) \nmid d$, $g^{-d}-1$ is a unit. Otherwise $v_p(g^{-d}-1) = e(d)$ by lifting the exponent. For the halves, $k \mapsto (2l+1)p^v - k$ exchanges them and preserves k^{-2} modulo p^v. □

Lemma 2.5 (block Wolstenholme; Beukers' Lemma 1)**.** *$\sigma_t(l) \equiv 0 \pmod{p^{2t}}$ for all $t, l \ge 0$. Hence $H'_x \equiv 0 \pmod{p^{2v}}$ whenever $p^v \mid x$.*

Proof. For $t \ge 1$, $k \mapsto (2l+1)p^t - k$ is a fixed-point-free involution of the units of $B_t(l)$, and $\frac{1}{k} + \frac{1}{(2l+1)p^t-k} \equiv -\frac{(2l+1)p^t}{k^2} \pmod{p^{2t}}$; apply Lemma 2.4. □

Lemma 2.6 (digit dropping)**.** *Let $m \in \mathbb{Z}$, $s \ge 1$, $L = mp^s$, $L' = L/p$, $n \ge 0$, $P_n = \prod'_{\lambda \le n}(1 - L/\lambda)$. Then exactly*

$$\binom{L-1}{n} = (-1)^{n-\lfloor n/p \rfloor}\binom{L'-1}{\lfloor n/p \rfloor}P_n, \qquad \binom{L}{n} = \binom{L'}{n/p}P_n \quad (p \mid n).$$

Consequently $(-1)^n\binom{mp^s-1}{n} \equiv (-1)^{\lfloor n/p \rfloor}\binom{mp^{s-1}-1}{\lfloor n/p \rfloor}$ and $\binom{mp^s}{pn} \equiv \binom{mp^{s-1}}{n} \pmod{p^s}$; also $\binom{mp^s+n}{n} \equiv \binom{mp^{s-1}+\lfloor n/p \rfloor}{\lfloor n/p \rfloor}$ and $\binom{j}{mp^s} \equiv \binom{\lfloor j/p \rfloor}{mp^{s-1}} \pmod{p^s}$.

Proof. Split $\prod_{\lambda \le n}(L-\lambda)/\lambda$ into $\lambda = p\mu$, whose product is $\binom{L'-1}{\lfloor n/p \rfloor}$, and $p \nmid \lambda$. Each of the latter factors is $\equiv -1 \pmod{p^s}$. The other statements follow by $m \mapsto -m$ and the identities of the notation section. □

3. The Dixon-type sum A361711

3.1. **Closed forms.**

Lemma 3.1. *For $m \geq 0$ and $m \geq 1$ respectively,*

$$a(2m+1) = (-1)^m \binom{3m+1}{m}\binom{2m}{m}, \qquad a(2m) = \frac{(-1)^m(2m-3)}{6(2m-1)}\binom{3m}{m}\binom{2m}{m}.$$

Proof. Put $F(n,k) = (-1)^k\binom{n}{k}^2\binom{n-2}{k}$, so that $a(n) = \sum_{k=0}^{n} F(n,k)$ for $n \geq 2$. Zeilberger's algorithm [19] produces the recurrence

$$c_0(n)F(n,k) + c_1(n)F(n+1,k) + c_2(n)F(n+2,k) = G(n,k+1) - G(n,k) \tag{4}$$

with $c_0 = -3n(n-1)(3n+1)(3n+2)(3n^2+4n+2)$, $c_1 = -6n(6n^3+12n^2+5n+1)$, $c_2 = -n^2(n+2)^2(3n^2-2n+1)$ and $G = RF$, $R(n,k) = \frac{k^3(n-1)P(n,k)}{(n-k-1)(n-k)(n-k+1)^2(n-k+2)^2}$, where P is the polynomial of Appendix A. Because R has poles at $k \in \{n-1,\ldots,n+2\}$, we do not divide by F; instead put

$$\Phi(n,k) = \frac{(-1)^k n!^2(n-2)!}{k!^3(n-k+2)!^2(n-k)!} \qquad (2 \leq n,\ 0 \leq k \leq n),$$

finite and nonzero. As exact identities of rational numbers for $0 \leq k \leq n$, $F(n,k) = \Phi\,(n-k+2)^2(n-k+1)^2(n-k)(n-k-1)$, $F(n+1,k) = \Phi\,(n+1)^2(n-1)(n-k+2)^2(n-k)$, $F(n+2,k) = \Phi\,(n+2)^2(n+1)^2n(n-1)$, $G(n,k) = \Phi\,k^3(n-1)P(n,k)$ and $G(n,k+1) = -\Phi\,(n-1)(n-k+2)^2(n-k)P(n,k+1)$ (with $G(n,n+1) := 0$). Thus (4) for $0 \leq k \leq n$ is Φ times a polynomial identity in (n,k), which is verified by expansion. Summing over $0 \leq k \leq n$, which covers the supports of $F(n,\cdot)$, $F(n+1,\cdot)$, $F(n+2,\cdot)$, gives $c_0a(n) + c_1a(n+1) + c_2a(n+2) = G(n,n+1) - G(n,0) = 0$ for $n \geq 2$. Both closed forms satisfy the same recurrence (a rational-function identity in n), $c_2(n) \neq 0$ for $n \geq 1$, and the initial values $a(2) = 1$, $a(3) = -8$ agree. □

3.2. **An exact product identity.**

Lemma 3.2. *Let p be an odd prime, n odd with $p \nmid n$, $k \geq 1$, $M = np^k$, $M' = np^{k-1}$, $\nu = (M-1)/2$, $\nu' = (M'-1)/2$ and U as above. Then*

$$\frac{\binom{2\nu}{\nu}}{\binom{2\nu'}{\nu'}} = (-1)^{\frac{p-1}{2}}\prod_{u\in U}\Big(1-\frac{M}{u}\Big), \qquad \frac{\binom{3\nu+1}{\nu}}{\binom{3\nu'+1}{\nu'}} = \prod_{u\in U}\Big(1+\frac{M}{u}\Big).$$

Proof. As $2\nu = M-1$ and $3\nu+1 = M+\nu$,

$$\binom{2\nu}{\nu} = (-1)^\nu\prod_{i=1}^{\nu}\Big(1-\frac{M}{i}\Big), \qquad \binom{M+\nu}{\nu} = \prod_{i=1}^{\nu}\Big(1+\frac{M}{i}\Big).$$

Write $i = p^a u$ with $p \nmid u$; the factor is $1 \mp p^{k-a}n/u$, and for fixed a the index u runs over $1 \leq u \leq \lfloor \nu/p^a \rfloor$. We claim $\lfloor \nu/p^a \rfloor = \lfloor \nu'/p^{a-1} \rfloor$ for $a \geq 1$. Indeed $\nu = p\nu' + \frac{p-1}{2}$; write $p\nu' = p^aQ + R$ with $0 \leq R < p^a$; then $p \mid R$, so $R \leq p^a - p$ and $R + \frac{p-1}{2} < p^a$, whence both floors equal Q. Consequently the factors with $a \geq 1$ at level k are exactly the factors of the corresponding product at level $k-1$, and the factors with $a = 0$ are those indexed by U. Finally $\nu - \nu' = \frac{p-1}{2}np^{k-1}$ has the parity of $\frac{p-1}{2}$. □

Proof of Theorem 1.1. (1) If $p \mid n$, write $n = p^e n'$ with $p \nmid n'$; the claim for (n,k) is weaker than the claim for $(n', k+e)$. So let $p \nmid n$. The arguments M and M' are odd, so Lemma 3.1 gives $a(M)/a(M')$ as $(-1)^{\nu-\nu'}$ times the product of the two quotients of Lemma 3.2; the signs

cancel and (2) follows. Expanding the product, $a(M)/a(M') = 1 - M^2H + M^4e_2 - M^6e_3 + \cdots$, where e_j are elementary symmetric functions of the u^{-2}, $u \in U$, and are p-integral. By Lemma 2.2, $v_p(M^2H) \geq 3k$, so $a(M)/a(M') \equiv 1 \pmod{p^{3k}}$. As $a(M')$ is a nonzero integer, $a(M) - a(M') = a(M')(a(M)/a(M') - 1) \equiv 0 \pmod{p^{3k}}$.

(2) $a(2) = 1$ and $a(10) = -98098$ by Lemma 3.1. □

Remark 3.3. (i) For $p = 3$ the statement fails: $v_3(H) = k - 1$ and $a(3) - a(1) = -9$. (ii) In the even case the factor $\frac{2m-3}{2m-1}$ of Lemma 3.1 does not scale, and in our computations ($p \in \{5, 7, 11, 13\}$, even $n \leq 12$, $k \leq 2$) the congruence fails in every case. (iii) Put $X = \prod_U (1 - M/u)$, $Y = \prod_U (1 + M/u)$ for $n = 1$. Then $XY \equiv 1 \pmod{p^{3k}}$, and Cai's prime-power form of Morley's congruence [7, 16] gives $X \equiv 4^{\varphi(p^k)}$; hence

$$\binom{(3p^k - 1)/2}{(p^k - 1)/2} \equiv 4^{-(p^k - p^{k-1})} \binom{(3p^{k-1} - 1)/2}{(p^{k-1} - 1)/2} \pmod{p^{3k}}.$$

This congruence is the case of modulus $n = p^k$ and parameter 3 (their $k = 3$) of Theorem 1.4 of Zhong, Chern and Cai [26]; the argument above gives a different proof of this case.

4. Integrality

4.1. **Integer forms.** Since $\Gamma(1 + x + m)/\Gamma(1 + x) = \prod_{j=1}^{m} (x + j)$,

$$\frac{\Gamma(1 + \frac{n}{2})}{\Gamma(1 + \frac{3n}{2})} = \frac{2^n}{\prod_{j=1}^{n}(n + 2j)}, \quad \frac{\Gamma(1 + \frac{n}{3})}{\Gamma(1 + \frac{4n}{3})} = \frac{3^n}{\prod_{j=1}^{n}(n + 3j)}, \quad \frac{\Gamma(1 + \frac{2n}{3})}{\Gamma(1 + \frac{5n}{3})} = \frac{3^n}{\prod_{j=1}^{n}(2n + 3j)},$$

so that

$$\begin{aligned} b(n) := \mathrm{A364178}(n) &= \frac{2^n (10n)!(3n)!}{(6n)!(5n)!\, n! \prod_{j \leq n}(n + 2j)}, \\ \mathrm{A364179}(n) &= \frac{2^n (10n)!}{(5n)!(4n)! \prod_{j \leq n}(n + 2j)}, \\ \mathrm{A364172}(n) &= \frac{3^n (6n)!}{(3n)!(2n)! \prod_{j \leq n}(n + 3j)}, \\ a(n) := \mathrm{A364175}(n) &= \frac{3^n (6n)!}{(3n)!(2n)! \prod_{j \leq n}(2n + 3j)}. \end{aligned}$$

4.2. **A Landau-type criterion with a correction term.** Let q be a prime, $d \in \{2, 3\}$ with $q \nmid d$, and $x \geq 1$. For $k \geq 1$ write $n = t_k q^k + \rho_k$ with $0 \leq \rho_k < q^k$. Since d is invertible modulo q^k, the $j \in [1, n]$ with $q^k \mid xn + dj$ form a single residue class modulo q^k; hence

$$N_k := \#\{1 \leq j \leq n : \ q^k \mid xn + dj\} = t_k + \varepsilon_k, \qquad \varepsilon_k \in \{0, 1\},$$

and $\varepsilon_k = 1$ only if $q^k \mid x\rho_k + ds$ for some $s \in [1, \rho_k]$. By Legendre's formula, for a ratio $\frac{d^n \prod_i (a_i n)!^{\epsilon_i}}{\prod_{j \leq n}(xn + dj)}$ with $\sum_i \epsilon_i a_i = 1$,

$$v_q\Big(\frac{d^n \prod_i (a_i n)!^{\epsilon_i}}{\prod_{j \leq n}(xn + dj)}\Big) = \sum_{k \geq 1} \Big(F(\rho_k/q^k) - \varepsilon_k \Big), \qquad F(y) := \sum_i \epsilon_i \lfloor a_i y \rfloor, \tag{5}$$

because $\left\lfloor a_i n/q^k \right\rfloor = a_i t_k + \left\lfloor a_i \rho_k/q^k \right\rfloor$ and the t_k cancel. This is Landau's criterion [14] with a correction ε_k coming from the linear factors.

Proof of Theorem 1.2. A364178 at odd q ($d=2$, $x=1$; here $\prod_i (a_i n)!^{\epsilon_i} = \frac{(10n)!(3n)!}{(6n)!(5n)!n!}$). Then $F(y) = \lfloor 10y\rfloor + \lfloor 3y\rfloor - \lfloor 6y\rfloor - \lfloor 5y\rfloor - \lfloor y\rfloor$ vanishes on $[0,\frac{1}{10}) \cup [\frac{1}{6},\frac{3}{10})$ and is ≥ 1 elsewhere on $[0,1)$. If $\varepsilon_k = 1$ then $0 < \rho_k + 2s \leq 3\rho_k < 3q^k$ forces $q^k \leq 3\rho_k$, i.e. $\rho_k/q^k \geq \frac{1}{3}$, where $F \geq 1$. Every summand of (5) is ≥ 0.

A364178 at q = 2. With $v_2(X!) = X - s_2(X)$: for odd n every $n+2j$ is odd and $v_2(b(n)) = 2n + s_2(n)$; for $n = 2m$, $\prod_{j\leq n}(n+2j) = 2^n(3m)!/m!$ and $v_2(b(n)) = s_2(3m)$.

A364172 at $q \neq 3$ ($d=3$, $x=1$, $F(y) = \lfloor 6y\rfloor - \lfloor 3y\rfloor - \lfloor 2y\rfloor$, which is 0 on $[0,\frac{1}{6})$ and ≥ 1 on $[\frac{1}{6},1)$). If $\varepsilon_k = 1$ then $0 < \rho_k + 3s \leq 4\rho_k < 4q^k$ forces $\rho_k/q^k \geq \frac{1}{4} > \frac{1}{6}$. This includes $q = 2$.

A364172 at q = 3. With $v_3(X!) = (X - s_3(X))/2$ and $s_3(3X) = s_3(X)$:

$$v_3\Big(\frac{3^n(6n)!}{(3n)!(2n)!}\Big) = \frac{3n + s_3(n)}{2}.$$

If $3 \nmid n$, every $n+3j$ is prime to 3. If $n = 3m$, $\prod_{j\leq n}(n+3j) = 3^n(4m)!/m!$, and $v_3(\text{A364172}(n)) = s_3(4m)/2 \geq 0$.

A364179 at odd q ($d=2$, $x=1$, $F(y) = \lfloor 10y\rfloor - \lfloor 5y\rfloor - \lfloor 4y\rfloor$, zero exactly on $[0,\frac{1}{10}) \cup [\frac{1}{4},\frac{3}{10})$). As for A364178, $\varepsilon_k = 1$ forces $\rho_k/q^k \geq \frac{1}{3} > \frac{3}{10}$, where $F \geq 1$.

A364179 at q = 2. For odd n, $v_2 = 2n + s_2(n)$; for $n = 2m$, $v_2 = s_2(3m)$. □

5. The unit-product principle and the towers

Let $p \geq 5$. For $X \geq 0$ put $U(X) = \prod'_{1\leq k\leq X} k$ and $W_r = U(p^r)$.

Lemma 5.1. *(1) $X! = p^{\lfloor X/p\rfloor}\, \lfloor X/p\rfloor!\, U(X)$ for every $X \geq 0$.*

(2) If $N = np^r$ and $c \geq 1$, then $U(cN) \equiv W_r^{cn} \pmod{p^{3r}}$.

(3) If $N = np^r$, $c \in \mathbb{Z}$ and $p \nmid d$, then $\prod'_{1\leq k\leq N}(1 + \frac{cN}{dk}) \equiv 1 \pmod{p^{3r}}$.

Proof. (1) Split the factors of $X!$ into multiples of p and units. (2) The units up to cN are $\beta p^r + u$ with $0 \leq \beta < cn$, $u \in R_r$, so $U(cN) = W_r^{cn} \prod_\beta P_\beta$ with $P_\beta = \prod_{u\in R_r}(1 + \beta p^r/u) = 1 + \beta p^r e_1 + \beta^2 p^{2r} e_2 + \cdots$, where the e_j are the elementary symmetric functions of the u^{-1}. By Lemma 2.1, $v_p(e_1) = v_p(S_1) \geq 2r$ and $v_p(e_2) = v_p((S_1^2 - S_2)/2) \geq r$; all terms beyond 1 are divisible by p^{3r}. (3) All factors lie in $1 + p\mathbb{Z}_p$, and log is an isometry there (p odd). The logarithm of the product is $\sum_{j\geq 1} \frac{(-1)^{j+1}}{j}(\frac{cN}{d})^j H^{(j)}$ with $H^{(j)} = \sum'_{k\leq N} k^{-j}$; by Lemma 2.1 the term $j = 1$ has valuation $\geq r + 2r$, the term $j = 2$ has valuation $\geq 2r + r$, and for $j \geq 3$ the valuation is $\geq jr - v_p(j) \geq 3r$. □

Theorem 5.2. *Let*

$$R(n) = \prod_i (a_i n)!^{\epsilon_i} \cdot \prod_t \frac{(x_t n/d_t)!}{(y_t n/d_t)!} \qquad (a_i, x_t, d_t \geq 1,\ \epsilon_i = \pm 1,\ y_t = x_t + c_t d_t,\ c_t \geq 1)$$

be balanced, $\sum_i \epsilon_i a_i = \sum_t c_t$. Then for every prime $p \geq 5$ with $p \nmid \prod_t d_t$ and all $n, r \geq 1$, $R(np^r)/R(np^{r-1}) \equiv 1 \pmod{p^{3r}}$.

Proof. Let $N = np^r$, $M = N/p$. Each quotient of fractional factorials equals $d_t^{c_t n}/\prod_{j=1}^{c_t n}(x_t n + d_t j)$. In $\prod_{j\leq c_t N}(x_t N + d_t j)$ the factors with $p \mid j = pj'$ are $p(x_t M + d_t j')$ and reproduce $p^{c_t M}\prod_{j'\leq c_t M}(x_t M + d_t j')$; the factors with $p \nmid j$ are $d_t j(1 + x_t N/(d_t j))$, with product $d_t^{c_t N - c_t M}\, U(c_t N) \prod'_{j\leq c_t N}(1 + \frac{x_t N}{d_t j})$. By Lemma 5.1(1) the integer factorials contribute

$p^{M\sum\epsilon_i a_i}\cdot(\text{level-}M\text{ value})\cdot\prod_i U(a_iN)^{\epsilon_i}$. The powers of p and of the d_t cancel by balance, so

$$\frac{R(N)}{R(M)}=\frac{\prod_i U(a_iN)^{\epsilon_i}}{\prod_t U(c_tN)}\prod_t\prod_{j\le c_tN}{}'\Big(1+\frac{x_tN}{d_tj}\Big)^{-1}.$$

By Lemma 5.1(2) the first factor is $\equiv W_r^{n(\sum\epsilon_i a_i-\sum c_t)}=1$, and by Lemma 5.1(3), applied with $c_tN=(c_tn)p^r$ in place of N, so is each product. □

Proof of Theorem 1.3. The four sequences are balanced ratios of the shape of Theorem 5.2 with $d_t\in\{2,3\}$, so $p\ge5$ is admissible. They are integers (Theorem 1.2 and [20]), hence $a(N)-a(M)=a(M)(a(N)/a(M)-1)\equiv0\pmod{p^{3r}}$. □

Remark 5.3. The proof shows $v_p(a(np^r))=v_p(a(n))$ for $p\ge5$, without any integrality assumption. For the other nine entries of A364172–A364184 Bala's conjecture is therefore equivalent to integrality, which the criterion of (5) does not settle for them (the correction terms are not dominated by F on an interval).

6. Proof of Theorem 1.4

Let $N=mp^r$, $M=N/p$, and assume $p\nmid m$ (the case $p\mid m$ follows from the case $(m/p^e,r+e)$). Write $a(n)=\sum_{k=1}^n T_k(n)$ with $T_k(n)=(-1)^{n+k}\binom{n}{k}\binom{n+k-1}{k}\binom{2k}{n}$ for $n\ge1$.

Lemma 6.1 (scaled terms)**.** *For $p\nmid k$ and $s\ge1$: $T_{kp^s}(N)\equiv T_{kp^{s-1}}(M)\pmod{p^{3r}}$.*

Proof. Put $K=kp^s$, $K'=K/p$. The signs agree since p is odd. The three quotients are $\binom{N}{K}/\binom{M}{K'}$, $\binom{N+K-1}{K}/\binom{M+K'-1}{K'}=\binom{N+K}{K}/\binom{M+K'}{K'}$ (the factors $\frac{N}{N+K}$ and $\frac{M}{M+K'}$ coincide), and $\binom{2K}{N}/\binom{2K'}{M}$. By Lemma 2.3 each is $\equiv1\pmod{p^{r+s+\min(r,s)}}$, and the vanishing conditions agree at both levels. If $s\ge r$ we are done. If $s<r$, the product is $\equiv1\pmod{p^{r+2s}}$, while $\binom{M}{K'}=\frac{M}{K'}\binom{M-1}{K'-1}$ and $\binom{M+K'-1}{K'}=\frac{M}{K'}\binom{M+K'-1}{K'-1}$ give $v_p(T_{K'}(M))\ge2(r-s)$. □

Summing over the scaled indices, $a(N)-a(M)\equiv G_0:=\sum_k{}' T_k(N)\pmod{p^{3r}}$. For $p\nmid k$ the two free factors N/k give

$$G_0=(-1)^N N^2S,\qquad S=\sum_k{}'\frac{(-1)^k}{k^2}\binom{N-1}{k-1}\binom{N+k-1}{k-1}\binom{2k}{N},$$

and it remains to show $S\equiv0\pmod{p^r}$. Put $D(n,k,j)=(-1)^k\binom{n-1}{k}\binom{n+k}{k}\binom{j}{n}$. Lemma 2.6, with $\lfloor(k-1)/p\rfloor=\lfloor k/p\rfloor$ for $p\nmid k$, gives $S\equiv-\Sigma_1\pmod{p^r}$, where

$$\Sigma_s=\sum_{1\le k\le N}{}' k^{-2}\,D\big(mp^{r-s},\lfloor k/p^s\rfloor,\lfloor 2k/p^s\rfloor\big)\qquad(1\le s\le r).$$

Lemma 6.2. *$\Sigma_s\equiv\Sigma_{s+1}\pmod{p^r}$ for $1\le s<r$, and $\Sigma_r\equiv0\pmod{p^r}$.*

Proof. Group k by $n=\lfloor k/p^s\rfloor$ and by the half of $B_s(n)$ containing k; this decides $\lfloor 2k/p^s\rfloor\in\{2n,2n+1\}$ (as p^s is odd). On each half-block the kernel D is constant and $\sum k^{-2}\equiv0\pmod{p^s}$ (Lemma 2.4), since N is a multiple of p^s and all blocks are complete. For $s=r$ this proves $\Sigma_r\equiv0$. For $s<r$, Lemma 2.6 at level $r-s$ gives $D(mp^{r-s},n,j)\equiv D(mp^{r-s-1},\lfloor n/p\rfloor,\lfloor j/p\rfloor)\pmod{p^{r-s}}$; the replacement costs $p^{r-s}\cdot p^s$, and since $\lfloor\lfloor k/p^s\rfloor/p\rfloor=\lfloor k/p^{s+1}\rfloor$ and $\lfloor\lfloor 2k/p^s\rfloor/p\rfloor=\lfloor 2k/p^{s+1}\rfloor$, the result is Σ_{s+1}. □

Hence $S\equiv0\pmod{p^r}$ and Theorem 1.4 follows. □

7. The first congruence of Theorem 1.5

Substituting $j = n-k$ and $\binom{2n-k-i}{n-k-i} = \binom{2n-k-i}{n}$,

$$A(n) = \sum_{k,i\geq 0} \binom{n}{k}^2 \binom{n+k}{k} \binom{n}{i}^2 \binom{2n-k-i}{n}.$$

Let $N = mp^r$, $M = N/p$, $p \nmid m$, and split the pairs (k,i) into classes S ($p \mid k$, $p \mid i$), K ($p \nmid k$, $p \mid i$), I ($p \mid k$, $p \nmid i$), B ($p \nmid k$, $p \nmid i$).

Class B carries $\binom{N}{k}^2\binom{N}{i}^2$, hence p^{4r}.

Class S. For $k = pK'$, $i = pI'$ the ratio of the terms at levels N and M is $Q_1^2Q_2Q_3^2Q_4$ with Kazandzidis quotients $Q_1 = \binom{N}{pK'}/\binom{M}{K'}$, $Q_2 = \binom{N+pK'}{pK'}/\binom{M+K'}{K'}$, $Q_3 = \binom{N}{pI'}/\binom{M}{I'}$, $Q_4 = \binom{p(2M-K'-I')}{pM}/\binom{2M-K'-I'}{M}$. With $\mu = \min(v_p(k), v_p(i))$, Lemma 2.3 gives every quotient $\equiv 1 \pmod{p^{r+2\mu}}$ if $\mu < r$ (and $\pmod{p^{3r}}$ if $\mu \geq r$). Also v_p of the level-M term is $\geq 2(r-\mu)$ by free valuation of $\binom{M}{K'}^2\binom{M}{I'}^2$. So $\Sigma_S \equiv A(M) \pmod{p^{3r}}$.

Class K. The factor $\binom{N}{k}^2 = \frac{N^2}{k^2}\binom{N-1}{k-1}^2$ gives $\Sigma_K = N^2 S_K$, and Lemma 2.6 together with Lemma 2.3 (in the form $\binom{N}{pi'} \equiv \binom{M}{i'} \pmod{p^r}$) gives

$$S_K \equiv \sum_k{}' k^{-2}\Psi_1(M, \lfloor k/p \rfloor) \pmod{p^r},$$

$$\Psi_1(M', n') = \binom{M'-1}{n'}^2 \binom{M'+n'}{n'} \sum_{i'} \binom{M'}{i'}^2 \binom{2M'-n'-1-i'}{M'},$$

using $\lfloor (2N-k-pi')/p \rfloor = 2M - i' - \lfloor k/p \rfloor - 1$ for $p \nmid k$.

Lemma 7.1 (self-similarity)**.** *For $t \geq 1$ and $n' \geq 0$: $\Psi_1(mp^t, n') \equiv \Psi_1(mp^{t-1}, \lfloor n'/p \rfloor) \pmod{p^t}$.*

Proof. The outer factors reduce by Lemma 2.6. In the inner sum the terms with $p \nmid i'$ carry $\binom{mp^t}{i'}^2 \equiv 0 \pmod{p^{2t}}$. For $i' = pi''$, Lemma 2.3 gives $\binom{mp^t}{pi''} \equiv \binom{mp^{t-1}}{i''}$ with exponent $\geq t+2$, and Lemma 2.6 gives $\binom{2mp^t-n'-1-pi''}{mp^t} \equiv \binom{2mp^{t-1}-i''-\lfloor n'/p \rfloor - 1}{mp^{t-1}}$, using $\lceil (n'+1)/p \rceil = \lfloor n'/p \rfloor + 1$. □

Now set $\Sigma_s = \sum_{k\leq N}{}' k^{-2}\Psi_1(mp^{r-s}, \lfloor k/p^s \rfloor)$. On every full block $B_s(n)$ the kernel is constant and the weights sum to $\equiv 0 \pmod{p^s}$; Lemma 7.1 turns Σ_s into Σ_{s+1} at cost $p^{r-s}p^s$, and $\Sigma_r \equiv 0$. Hence $S_K \equiv 0 \pmod{p^r}$ and $\Sigma_K \equiv 0 \pmod{p^{3r}}$. *Class I* is identical, with the kernel $\Psi_2(M', n') = \binom{M'-1}{n'}^2 \sum_{k'} \binom{M'}{k'}^2 \binom{M'+k'}{k'} \binom{2M'-k'-n'-1}{M'}$. This proves the first congruence. □

8. The shifted congruence

8.1. **Set-up.** With $j = n-k-i$, the summand becomes $\binom{n}{k}^2\binom{n+k}{k}\binom{n}{k+j}^2\binom{n+j}{j}$, so for $N \geq 1$

$$A(N-1) = \sum_{k,j\geq 0} \tilde{S}(N;k,j), \qquad \tilde{S}(N;k,j) = \binom{N-1}{k}^2 \binom{N+k-1}{k} \binom{N-1}{k+j}^2 \binom{N+j-1}{j}.$$

Let $N = mp^r$ with $m, r \geq 1$, $M = N/p$ and $L_u = mp^u$ ($u \geq 0$). In this section "integral" means "in $\mathbb{Z}_p$", and we do not assume $p \nmid m$. Split the pairs (k,j) into the classes S ($p \mid k$, $p \mid j$), K ($p \nmid k$, $p \mid j$), J ($p \mid k$, $p \nmid j$) and B ($p \nmid k$, $p \nmid j$), and let $\Sigma_S, \Sigma_K, \Sigma_J, \Sigma_B$ be the corresponding partial sums of $\tilde{S}(N;k,j)$. We prove $\Sigma_S \equiv A(M-1)$ and $\Sigma_K \equiv \Sigma_J \equiv \Sigma_B \equiv 0 \pmod{p^{3r}}$. Here $\binom{N-1}{\cdot}$ carries no free power of p, so classes K and J have only one explicit factor N and require a cancellation modulo p^{2r}.

8.2. **A first-order expansion.**

Lemma 8.1 (first-order expansion)**.** *Let $L \in p\mathbb{Z}_p$ and, for $x \geq 0$, $P_x := \prod'_{\lambda \leq x}(1 - L/\lambda)$ and $\bar{P}_x := \prod'_{\lambda \leq x}(1 + L/\lambda)$. Let $E = \prod_i P_{x_i}^{a_i} \bar{P}_{y_i}^{b_i}$ be a finite product with $a_i, b_i \geq 0$. Then $E = \sum_{d \geq 0} L^d e_d$ with $e_0 = 1$, every e_d integral,*

$$e_1 = -\sum_i a_i H'_{x_i} + \sum_i b_i H'_{y_i}, \qquad e_2 = \tfrac{1}{2}e_1^2 - \tfrac{1}{2}\sum_i \big(a_i S_2(x_i) + b_i S_2(y_i)\big).$$

Proof. Group the factors by λ. The factor belonging to λ is $(1-y)^{\alpha_\lambda}(1+y)^{\beta_\lambda}$ with $y = L/\lambda$, where $\alpha_\lambda, \beta_\lambda \geq 0$ count the products P_{x_i}, $\bar{P}_{y_i}$ (with multiplicity) whose range contains λ. Its expansion is $1 + c_\lambda y + d_\lambda y^2 + \cdots$ with integer coefficients, $c_\lambda = \beta_\lambda - \alpha_\lambda$ and $d_\lambda = \binom{\alpha_\lambda}{2} + \binom{\beta_\lambda}{2} - \alpha_\lambda\beta_\lambda$. Hence e_d is a $\mathbb{Z}$-combination of products of d factors $1/\lambda$ with $p \nmid \lambda$, so it is integral, and $e_1 = \sum_\lambda c_\lambda/\lambda$ is the stated expression. Finally $e_2 = \sum_{\lambda<\mu} \frac{c_\lambda c_\mu}{\lambda\mu} + \sum_\lambda \frac{d_\lambda}{\lambda^2} = \frac{1}{2}e_1^2 + \sum_\lambda \frac{d_\lambda - c_\lambda^2/2}{\lambda^2}$ and $d_\lambda - \frac{1}{2}c_\lambda^2 = -\frac{1}{2}(\alpha_\lambda + \beta_\lambda)$. □

With $L = mp^s$, the products P_n are those of Lemma 2.6, and replacing m by $-m$ there gives the analogous identities with $\bar{P}_n$.

8.3. **Harmonic lemmas.**

Lemma 8.2 (reflection)**.** *Let $t \geq 1$. For $0 \leq n < p^t$,*

$$H'_{p^t-1-n} - H'_n \equiv -p^t \sum_{n<\mu<p^t}{}' \mu^{-2} \pmod{p^{2t}};$$

in particular $H'_{p^t-1-n} \equiv H'_n \pmod{p^t}$. For $s \geq 1$ and $0 \leq k \leq p^t$, $H'_{(p^t-k)p^s} \equiv H'_{kp^s} \pmod{p^{t+2s}}$.

Proof. With $\mu = p^t - \lambda$,

$$H'_{p^t-1-n} = \sum_{n<\mu<p^t}{}' \frac{1}{p^t - \mu} = -\sum_{n<\mu<p^t}{}' \frac{1}{\mu}\Big(1 - \frac{p^t}{\mu}\Big)^{-1} = -(H'_{p^t-1} - H'_n) - p^t \sum_{n<\mu<p^t}{}' \mu^{-2} + p^{2t}(\text{integral}),$$

and $H'_{p^t-1} = H'_{p^t} \equiv 0 \pmod{p^{2t}}$ by Lemma 2.5. The same computation with $\mu = p^{t+s} - \lambda$ gives

$$H'_{(p^t-k)p^s} = -(H'_{p^{t+s}} - H'_{kp^s}) - p^{t+s} \sum_{kp^s<\mu<p^{t+s}}{}' \mu^{-2} + p^{2(t+s)}(\text{integral}).$$

The middle sum runs over full blocks of length p^s, so it is $\equiv 0 \pmod{p^s}$ by Lemma 2.4, and $H'_{p^{t+s}} \equiv 0 \pmod{p^{2t+2s}}$ by Lemma 2.5. □

Lemma 8.3 (block shift)**.** *Let $t \geq 1$, $i \geq 0$ and $n = ip^t + \bar{n}$ with $0 \leq \bar{n} < p^t$. Then $H'_n \equiv H'_{ip^t} + H'_{\bar{n}} \pmod{p^t}$, and $(ip^t + \lambda)^{-d} \equiv \lambda^{-d} \pmod{p^t}$ for $p \nmid \lambda$ and $d \geq 1$.*

Proof. $H'_n - H'_{ip^t} = \sum'_{\lambda \leq \bar{n}} (ip^t + \lambda)^{-1}$, and $(ip^t + \lambda)^{-1} \equiv \lambda^{-1} \pmod{p^t}$. □

Lemma 8.4 (Beukers' Lemma 4(i))**.** *For $s, t, i \geq 0$: $\sum_{n \in B_t(i)} \sigma_s(n) H'_n \equiv 0 \pmod{p^{t+2s}}$.*

Proof. For $t = 0$ the sum is $\sigma_s(i) H'_i \equiv 0 \pmod{p^{2s}}$ by Lemma 2.5. Let $t \geq 1$ and $t' = t + s$. Interchanging the summations, the sum equals $\Sigma(i) := \sum'_{k \in B_{t'}(i)} \frac{1}{k} H'_{\lfloor k/p^s \rfloor}$.

Base block $i = 0$, $s \geq 1$. The involution $k \mapsto p^{t'} - k$ of the units below $p^{t'}$ satisfies $\lfloor (p^{t'} - k)/p^s \rfloor = p^t - 1 - \lfloor k/p^s \rfloor$, since $p^s \nmid k$. With $n(k) = \lfloor k/p^s \rfloor$,

$$2\Sigma(0) = \sum_{k<p^{t'}}{}' \Big(\frac{1}{k} + \frac{1}{p^{t'}-k}\Big) H'_{n(k)} + \sum_{k<p^{t'}}{}' \frac{H'_{p^t-1-n(k)} - H'_{n(k)}}{p^{t'}-k}.$$

In the first sum, $\frac{1}{k} + \frac{1}{p^{t'}-k} \equiv -p^{t'}k^{-2} \pmod{p^{2t'}}$ and $\sum'_k k^{-2} H'_{n(k)} = \sum_{n<p^t} H'_n \sum'_{k \in B_s(n)} k^{-2} \equiv 0 \pmod{p^s}$ by Lemma 2.4, so the first sum is $\equiv 0 \pmod{p^{t+2s}}$. In the second sum, group k by $n = n(k)$: the numerator is p^t times an integer by Lemma 8.2, and $\sum'_{k \in B_s(n)} \frac{1}{p^{t'}-k} = \sigma_s(p^t - 1 - n) \equiv 0 \pmod{p^{2s}}$ by Lemma 2.5. As 2 is a unit, $\Sigma(0) \equiv 0 \pmod{p^{t+2s}}$.

Base block, $s = 0$. Here $n(k) = k$ and $H'_{p^t-k} = H'_{p^t-1-k} + \frac{1}{p^t-k}$, so

$$2\sum_{k<p^t}{}' \frac{H'_k}{k} = \sum_{k<p^t}{}' \frac{p^t H'_k}{k(p^t-k)} + \sum_{k<p^t}{}' \frac{H'_{p^t-1-k} - H'_k}{p^t-k} + \sum_{k<p^t}{}' \frac{1}{(p^t-k)^2}.$$

The three sums are $\equiv 0 \pmod{p^t}$: the first has the factor p^t, the second by Lemma 8.2, the third by Lemma 2.4.

General block, $i \geq 1$. Write $n = ip^t + \bar{n}$ with $0 \leq \bar{n} < p^t$. Exactly, with integral η, θ,

$$H'_{ip^t+\bar{n}} = H'_{ip^t} + H'_{\bar{n}} - ip^t S_2(\bar{n}) + p^{2t}\eta(\bar{n}),$$
$$\sigma_s(ip^t + \bar{n}) = \sigma_s(\bar{n}) - ip^{t'}\tau_s(\bar{n}) + p^{2t'}\theta(\bar{n}), \qquad \tau_s(\bar{n}) := \sum_{k \in B_s(\bar{n})}{}' k^{-2},$$

from $\frac{1}{x+\lambda} = \frac{1}{\lambda} - \frac{x}{\lambda^2} + \frac{x^2}{\lambda^2(x+\lambda)}$. Here $\tau_s(\bar{n}) \equiv 0 \pmod{p^s}$ (Lemma 2.4; empty for $s = 0$) and $\sigma_s(ip^t + \bar{n}) \equiv 0 \pmod{p^{2s}}$ (Lemma 2.5). Multiply the first expansion by $\sigma_s(ip^t + \bar{n})$ and sum over $\bar{n} < p^t$. The term with H'_{ip^t} gives $H'_{ip^t}\sigma_{t'}(i) \equiv 0 \pmod{p^{2t+2t'}}$. The term with $H'_{\bar{n}}$ gives, by the second expansion, $\Sigma(0) - ip^{t'}\sum_{\bar{n}} \tau_s(\bar{n}) H'_{\bar{n}} + p^{2t'}(\cdots) \equiv 0 \pmod{p^{t+2s}}$. The terms with $ip^t S_2(\bar{n})$ and $p^{2t}\eta$ carry p^t and p^{2t} times $\sigma_s \equiv 0 \pmod{p^{2s}}$. Hence $\Sigma(i) \equiv 0 \pmod{p^{t+2s}}$. □

Lemma 8.5 (Beukers' Lemma 4(ii), strong form)**.** *For* $s, t, i \geq 0$,

$$\sum_{k \in B_t(i)}{}' \frac{1}{k} H'_{kp^s} \equiv 0 \pmod{p^{t+2s}}.$$

Proof. For $s = 0$ this is Lemma 8.4 with $s = 0$; for $t = 0$ it follows from $H'_{kp^s} \equiv 0 \pmod{p^{2s}}$ (Lemma 2.5). Let $s, t \geq 1$. For the block $i = 0$, the involution $k \mapsto p^t - k$ gives

$$2\sum_{k<p^t}{}' \frac{H'_{kp^s}}{k} = \sum_{k<p^t}{}' \frac{p^t H'_{kp^s}}{k(p^t-k)} + \sum_{k<p^t}{}' \frac{H'_{(p^t-k)p^s} - H'_{kp^s}}{p^t-k};$$

the first sum is $\equiv 0 \pmod{p^{t+2s}}$ because $H'_{kp^s} \equiv 0 \pmod{p^{2s}}$, and the second by Lemma 8.2. For a general block write $k = ip^t + \bar{k}$; then $\frac{1}{k} = \frac{1}{\bar{k}} - \frac{ip^t}{\bar{k}^2} + p^{2t}(\text{integral})$ and

$$H'_{kp^s} = H'_{ip^{t+s}} + H'_{\bar{k}p^s} - ip^{t+s}S_2(\bar{k}p^s) + p^{2(t+s)}(\text{integral}),$$

with $H'_{ip^{t+s}} \equiv 0 \pmod{p^{2(t+s)}}$ and $S_2(\bar{k}p^s) \equiv 0 \pmod{p^s}$ (Lemmas 2.5 and 2.4). Multiplying out, every term other than the base-block sum $\sum'_{\bar{k}} \frac{1}{\bar{k}} H'_{\bar{k}p^s}$ is $\equiv 0 \pmod{p^{t+2s}}$: the corrections to $1/k$ carry p^t and meet $H'_{\bar{k}p^s} \equiv 0 \pmod{p^{2s}}$, and the correction $-ip^{t+s}S_2$ carries p^{t+2s}. □

Lemma 8.6 (carry-restricted sums)**.** *For $t \geq 1$, $i, i' \geq 0$ and $c \in \{0,1\}$, with bars denoting residues modulo p^t, let*

$$\mathrm{Cell}_t(i,i';c) := \{(n,j) : n \in B_t(i),\, j \in B_t(i'),\, p \nmid j,\, [\bar{n} + \bar{j} \geq p^t] = c\}.$$

Then

$$(Y) \quad \sum_{(n,j)\in \mathrm{Cell}_t(i,i';c)} \frac{\sigma_{s'}(n)}{j} \equiv 0 \pmod{p^{2s'+t}} \ \ (s' \geq 0), \qquad (B) \quad \sum_{\substack{(n,j)\in \mathrm{Cell}_t(i,i';c)\\ p\nmid n}} \frac{1}{nj} \equiv 0 \pmod{p^t}.$$

The condition $[\bar{n} + \bar{j} \geq p^t] = c$ is symmetric in n and j, so both statements also hold with the roles of the two indices exchanged.

Proof. (Y) The sum over both values of c is $\sigma_{s'+t}(i)\,\sigma_t(i') \equiv 0 \pmod{p^{2s'+4t}}$ by Lemma 2.5, so it suffices to take $c = 1$, that is $\bar{j} \geq p^t - \bar{n}$. By Lemma 8.3, $1/j = 1/\bar{j} + p^t(\text{integral})$, and the correction contributes $\equiv 0 \pmod{p^{t+2s'}}$ because $\sigma_{s'}(n) \equiv 0 \pmod{p^{2s'}}$. The main part is

$$\sum_{n\in B_t(i)} \sigma_{s'}(n)\big(H'_{p^t-1} - H'_{p^t-1-\bar{n}}\big).$$

Here $H'_{p^t-1}\sum_n \sigma_{s'}(n) = H'_{p^t-1}\sigma_{s'+t}(i) \equiv 0 \pmod{p^{2s'+4t}}$, and $H'_{p^t-1-\bar{n}} = H'_n - H'_{ip^t} + p^t(\text{integral})$ by Lemmas 8.2 and 8.3; the corrections again meet $\sigma_{s'}(n) \equiv 0 \pmod{p^{2s'}}$. What remains is $-\sum_{n\in B_t(i)} \sigma_{s'}(n)H'_n + H'_{ip^t}\sigma_{s'+t}(i) \equiv 0 \pmod{p^{2s'+t}}$ by Lemmas 8.4 and 2.5. (B) Both values of c together give $\sigma_t(i)\sigma_t(i') \equiv 0 \pmod{p^{4t}}$; take $c = 1$. Replacing $1/n$, $1/j$ by $1/\bar{n}$, $1/\bar{j}$ costs multiples of p^t, and

$$\sum_{\bar{n}<p^t}{}' \frac{1}{\bar{n}} \sum_{p^t-\bar{n}\leq\bar{j}<p^t}{}' \frac{1}{\bar{j}} = \sum_{\bar{n}<p^t}{}' \frac{H'_{p^t-1} - H'_{p^t-1-\bar{n}}}{\bar{n}} \equiv -\sum_{\bar{n}<p^t}{}' \frac{H'_{\bar{n}}}{\bar{n}} \equiv 0 \pmod{p^t}$$

by Lemma 8.2 and Lemma 8.5 with $s = 0$, $i = 0$. □

8.4. **Block inductions.** A family $(T_u)_{u\geq 0}$ of functions $\mathbb{Z}_{\geq 0} \to \mathbb{Z}_p$ is a *tower* if $T_u(k) \equiv T_{u-1}(\lfloor k/p \rfloor) \pmod{p^u}$ for all $u \geq 1$ and $k \geq 0$. The functions may change their form from level to level; only this property is used.

Lemma 8.7 (one-index block induction; Beukers' Lemma 3, Straub's Lemma 5.6)**.** *Let (T_u) be a tower and let $a_k \in \mathbb{Z}_p$ satisfy $\sum_{k\in B_t(l)} a_k \equiv 0 \pmod{p^t}$ for all $t, l \geq 0$. Then $\sum_{k\in B_s(l)} a_k T_s(k) \equiv 0 \pmod{p^s}$ for all $s, l \geq 0$, and hence $\sum_{k<mp^s} a_k T_s(k) \equiv 0 \pmod{p^s}$. If $a_k = 0$ whenever $p \mid k$, the tower property at the top level s is needed only for $p \nmid k$.*

Proof. Induction on s; $s = 0$ is trivial. By the tower property,

$$\sum_{k\in B_s(l)} a_k T_s(k) \equiv \sum_{q\in B_{s-1}(l)} \Big(\sum_{k\in B_1(q)} a_k\Big) T_{s-1}(q) \pmod{p^s}.$$

The numbers $\bar{a}_q := p^{-1}\sum_{k\in B_1(q)} a_k$ are integral, and $\sum_{q\in B_t(l')} \bar{a}_q = p^{-1}\sum_{k\in B_{t+1}(l')} a_k \equiv 0 \pmod{p^t}$, so the induction hypothesis applies to them. For the last statement, redefine $T_s(k) := T_{s-1}(\lfloor k/p \rfloor)$ on the multiples of p; this does not change the sum. □

Lemma 8.8 (two-index block induction with carries)**.** *Let $T_u^{(\delta)} : \mathbb{Z}_{\geq 0}^2 \to \mathbb{Z}_p$ ($u \geq 0$, $\delta \in \{0,1\}$) satisfy*

$$T_u^{(\delta)}(q,q') \equiv T_{u-1}^{(\delta')}\big(\lfloor q/p \rfloor, \lfloor q'/p \rfloor\big) \pmod{p^u}, \qquad \delta' := [(q \bmod p) + (q' \bmod p) + \delta \geq p].$$

For $t \geq 1$ let the cells be $\{(q,q',\delta) : q \in B_t(i),\ q' \in B_t(i'),\ [(q \bmod p^t) + (q' \bmod p^t) + \delta \geq p^t] = c\}$, and let $a^{(\delta)}_{q,q'} \in \mathbb{Z}_p$ have cell sums $\equiv 0 \pmod{p^t}$ for all $t \geq 1$, $i, i' \geq 0$ and $c \in \{0,1\}$. Then $\sum_{q \in B_s(l),\, q' \in B_s(l'),\, \delta} a^{(\delta)}_{q,q'} T^{(\delta)}_s(q,q') \equiv 0 \pmod{p^s}$ for all $s, l, l' \geq 0$.

Proof. As for Lemma 8.7, now collecting the terms by $(\lfloor q/p \rfloor, \lfloor q'/p \rfloor, \delta')$; each coefficient is a cell sum at level 1. The carries compose:

$$(q \bmod p^t) + (q' \bmod p^t) + \delta \\ = p[(q_1 \bmod p^{t-1}) + (q_1' \bmod p^{t-1}) + \delta'] + ((q \bmod p) + (q' \bmod p) + \delta - p\delta')$$

with $q_1 = \lfloor q/p \rfloor$, $q_1' = \lfloor q'/p \rfloor$ and the last bracket in $[0, p-1]$. So the level-t carry of (q,q',δ) is the level-$(t-1)$ carry of (q_1, q_1', δ'), and the collected weights satisfy the hypothesis one level lower. □

In the applications the top-level weights carry the label $\delta = 0$ only and vanish unless one index is prime to p. With these restrictions the top-level cells of Lemma 8.8 are the cells of Lemma 8.6, and, as in Lemma 8.7, the tower property at the top level is needed only where the weights are nonzero. The lower levels use the full tower property, which holds in every application.

8.5. **Kernels.** For $u \geq 0$, $n, j, q, q' \geq 0$ and $\delta \in \{0,1\}$ put

$$\Psi_u(n,j) := \binom{L_u - 1}{n}^2 (-1)^n \binom{-L_u - 1}{n} \binom{L_u - 1}{n+j}^2 (-1)^j \binom{-L_u}{j}, \qquad K_u(n) := \sum_{j \geq 0} \Psi_u(n,j),$$

$$\Xi^{(\delta)}_u(q,q') := \binom{L_u - 1}{q}^2 (-1)^q \binom{-L_u - 1}{q} \binom{L_u - 1}{q + q' + \delta}^2 (-1)^{q'} \binom{-L_u - 1}{q'},$$

$$\hat{\Xi}_u(n,j) := \binom{L_u - 1}{n}^2 (-1)^n \binom{-L_u - 1}{n} \binom{L_u - 1}{n+j}^2 (-1)^{j-1} \binom{-L_u - 1}{j-1} \quad (j \geq 1).$$

All are integers, and $\Psi_u(n,j) = 0$ unless $n + j < L_u$, so $K_u(n)$ is a finite sum.

Lemma 8.9 (how the terms factor)**.** *$\tilde{S}(N;k,j) = \frac{N}{N+k} \Psi_r(k,j)$ for all $k, j \geq 0$, and $\Psi_u(n,j) = \frac{L_u}{j} \hat{\Xi}_u(n,j)$ for $p \nmid j$.*

Proof. $\binom{N+k-1}{k} = (-1)^k \binom{-N}{k}$ and $\binom{-N}{k} = \binom{-N-1}{k} \frac{N}{N+k}$, from $\binom{a}{k} = \binom{a-1}{k} \frac{a}{a-k}$; likewise $\binom{N+j-1}{j} = (-1)^j \binom{-N}{j}$. For the second claim, $(-1)^j \binom{-L}{j} = \frac{L}{j} (-1)^{j-1} \binom{-L-1}{j-1}$. □

Lemma 8.10 (towers)**.** *Let $u \geq 1$.*

(1) $\Psi_u(q,j) \equiv 0 \pmod{p^u}$ *for $p \nmid j$, and* $\Psi_u(q, pj_1) \equiv \Psi_{u-1}(\lfloor q/p \rfloor, j_1) \pmod{p^u}$. *Hence (K_u) is a tower.*

(2) $\Xi^{(\delta)}_u(q,q') \equiv \Xi^{(\delta')}_{u-1}(\lfloor q/p \rfloor, \lfloor q'/p \rfloor) \pmod{p^u}$, *with δ' as in Lemma 8.8. For $p \nmid j$,* $\hat{\Xi}_u(n,j) \equiv \Xi^{(\delta)}_{u-1}(\lfloor n/p \rfloor, \lfloor j/p \rfloor) \pmod{p^u}$ *with* $\delta = [(n \bmod p) + (j \bmod p) \geq p]$.

(3) (moments) $K^{[t]}_u(q) := \sum_j j^t \Psi_u(q,j) \equiv 0 \pmod{p^u}$ *for $t \geq 1$ (and all $u \geq 0$).*

(4) (harmonic weights) *For $d \geq 1$,* $\tilde{K}^{(d)}_u(q) := \sum_j \Psi_u(q,j) H'_{p^d j}$ *satisfies*

$$\tilde{K}^{(d)}_u(q) \equiv \tilde{K}^{(d+1)}_{u-1}(\lfloor q/p \rfloor) \pmod{p^u}.$$

So for $s_0 \geq 0$, $d_0 \geq 1$ the functions $T_u := \tilde{K}^{(d_0 + s_0 - u)}_u$ $(0 \leq u \leq s_0)$ form a tower.

Proof. Put $L = L_u$, $L' = L_{u-1}$. (1) For $p \nmid j$ the last factor of Ψ_u is $\frac{L}{j}(-1)^{j-1}\binom{-L-1}{j-1}$, divisible by p^u. For $j = pj_1$, Lemma 2.6 with m and with $-m$ gives, modulo p^u, $\binom{L-1}{q}^2 \equiv \binom{L'-1}{\lfloor q/p \rfloor}^2$, $(-1)^q\binom{-L-1}{q} \equiv (-1)^{\lfloor q/p \rfloor}\binom{-L'-1}{\lfloor q/p \rfloor}$, $\binom{L-1}{q+pj_1}^2 \equiv \binom{L'-1}{\lfloor q/p \rfloor + j_1}^2$ and $(-1)^{pj_1}\binom{-L}{pj_1} \equiv (-1)^{j_1}\binom{-L'}{j_1}$. Summing over j gives the tower property of K_u. (2) The same factor-by-factor reduction, with $\lfloor (q+q'+\delta)/p \rfloor = \lfloor q/p \rfloor + \lfloor q'/p \rfloor + \delta'$. For $\hat{\Xi}_u$ use in addition $(-1)^{j-1}\binom{-L-1}{j-1} \equiv (-1)^{\lfloor j/p \rfloor}\binom{-L'-1}{\lfloor j/p \rfloor}$, which is Lemma 2.6 with $\lfloor (j-1)/p \rfloor = \lfloor j/p \rfloor$ for $p \nmid j$, and $\lfloor (n+j)/p \rfloor = \lfloor n/p \rfloor + \lfloor j/p \rfloor + \delta$. (3) Induction on u, the case $u = 0$ being empty. The unscaled j contribute $\equiv 0 \pmod{p^u}$ by (1), and the scaled $j = pj_1$ contribute $p^t \sum_{j_1} j_1^t \Psi_u(q, pj_1) \equiv p^t K_{u-1}^{[t]}(\lfloor q/p \rfloor) \pmod{p^{u+t}}$, which is divisible by p^{t+u-1}. (4) The unscaled j vanish modulo p^u by (1), and for $j = pj_1$, $H'_{p^d \cdot pj_1} = H'_{p^{d+1} j_1}$. □

Lemma 8.11 (first-order model)**.** *Let $u \geq 0$, $L = L_{u+1}$ and $P_x, \bar{P}_x$ as in Lemma 8.1. Then exactly*

$$\Psi_{u+1}(n, pj_1) = \Psi_u(\lfloor n/p \rfloor, j_1)\, P_n^2 \bar{P}_n P_{n+pj_1}^2 \bar{P}_{pj_1},$$

and hence $\Psi_{u+1}(n, pj_1) \equiv \Psi_u(\lfloor n/p \rfloor, j_1)(1 - L\rho_u(n, j_1)) \pmod{p^{2u+2}}$ with $\rho_u(n, j_1) := H'_n - H'_{pj_1} + 2H'_{n+pj_1}$.

Proof. With $q = \lfloor n/p \rfloor$ and $L' = L_u$, the exact identities of Lemma 2.6 (with m and $-m$) give $\binom{L-1}{n} = (-1)^{n-q}\binom{L'-1}{q}P_n$, $(-1)^n\binom{-L-1}{n} = (-1)^q\binom{-L'-1}{q}\bar{P}_n$, $\binom{L-1}{n+pj_1}^2 = \binom{L'-1}{q+j_1}^2 P_{n+pj_1}^2$ and $(-1)^{pj_1}\binom{-L}{pj_1} = (-1)^{j_1}\binom{-L'}{j_1}\bar{P}_{pj_1}$. Multiply. The congruence is Lemma 8.1 with $e_1 = -2H'_n + H'_n - 2H'_{n+pj_1} + H'_{pj_1} = -\rho_u(n, j_1)$ and $L^2 \in p^{2u+2}\mathbb{Z}_p$. □

8.6. **Class K.** Put $\hat{K}_r(k) := \sum_{j_1 \geq 0} \Psi_r(k, pj_1)$. By Lemma 8.9, for $p \nmid k$ the class-K terms with this k sum to $\frac{N}{N+k}\hat{K}_r(k) = (\frac{N}{k} - \frac{N^2}{k^2} + \frac{N^3}{k^3}(1 + \frac{N}{k})^{-1})\hat{K}_r(k)$, so

$$\Sigma_K \equiv N \sum_{k<N}{}' \frac{\hat{K}_r(k)}{k} - N^2 \sum_{k<N}{}' \frac{\hat{K}_r(k)}{k^2} \pmod{p^{3r}}. \tag{6}$$

The term with $1/k^2$. By Lemma 8.10(1) we have $\hat{K}_r(k) \equiv K_{r-1}(\lfloor k/p \rfloor)$ modulo p^r, so $(\hat{K}_r, K_{r-1}, K_{r-2}, \dots)$ is a tower. The weights k^{-2} ($p \nmid k$; 0 otherwise) have block sums $\equiv 0 \pmod{p^t}$ by Lemma 2.4, and Lemma 8.7 gives $\sum'_{k<N} k^{-2}\hat{K}_r(k) \equiv 0 \pmod{p^r}$.

The term with $1/k$. For $0 \leq s \leq r-1$ put $A_s := \sum_{n<L_s} \sigma_{r-s}(n) K_s(n)$. By Lemma 8.11 with $u = r-1$, $\hat{K}_r(k) \equiv \sum_{j_1} \Psi_{r-1}(\lfloor k/p \rfloor, j_1)(1 - N\rho_{r-1}(k, j_1)) \pmod{p^{2r}}$. Grouping k by $\lfloor k/p \rfloor$ and using $\sum'_{k \in B_1(n')} 1/k = \sigma_1(n')$,

$$\sum_{k<N}{}' \frac{\hat{K}_r(k)}{k} \equiv A_{r-1} - N\hat{X} \pmod{p^{2r}}, \qquad \hat{X} := \sum_{k<N}{}' \frac{1}{k} \sum_{j_1} \Psi_{r-1}(\lfloor k/p \rfloor, j_1)\rho_{r-1}(k, j_1).$$

So it suffices to show $\hat{X} \equiv 0 \pmod{p^r}$ and $A_{r-1} \equiv 0 \pmod{p^{2r}}$.

Proof that $\hat{X} \equiv 0 \pmod{p^r}$. Write $n' = \lfloor k/p \rfloor$ and split $\hat{X} = \hat{X}_1 - \hat{X}_2 + 2\hat{X}_3$ according to the three terms of ρ_{r-1}.

- $\hat{X}_1 = \sum_{n'} c_{n'} K_{r-1}(n')$ with $c_{n'} = \sum'_{k \in B_1(n')} H'_k/k$. By Lemma 8.5 ($s = 0$, level $t+1$), $\sum_{n' \in B_t(i)} c_{n'} \equiv 0 \pmod{p^{t+1}}$. So the weights $c_{n'}/p$ are integral with block sums $\equiv 0 \pmod{p^t}$, and Lemma 8.7 with the tower (K_u) gives $\hat{X}_1 \equiv 0 \pmod{p^r}$.

- $\hat{X}_2 = \sum_{n'} \sigma_1(n')\tilde{K}^{(1)}_{r-1}(n')$. The weights $\sigma_1(n')/p^2$ are integral, with block sums $\sigma_{t+1}(i)/p^2$ divisible by p^{2t} (Lemma 2.5), so Lemma 8.7 with the tower of Lemma 8.10(4) gives $\hat{X}_2 \equiv 0 \pmod{p^{r+1}}$.
- For $\hat{X}_3$, exactly $H'_{k+pj_1} = H'_{pj_1} + \sum'_{\lambda\le k} \frac{1}{pj_1+\lambda}$, and $\frac{1}{pj_1+\lambda} = \sum_{t\ge0}(-pj_1)^t\lambda^{-t-1}$ in $\mathbb{Z}_p$. Hence $\hat{X}_3 = \hat{X}_2 + \sum_{t\ge0}(-p)^t \sum'_k \frac{1}{k} S_{t+1}(k) K^{[t]}_{r-1}(n')$ with $K^{[0]} = K$. The term $t = 0$ is $\hat{X}_1$, and for $t \ge 1$ the term is $\equiv 0 \pmod{p^{t+r-1}}$ by Lemma 8.10(3). So $\hat{X}_3 \equiv 0 \pmod{p^r}$.

Proof that $A_{r-1} \equiv 0 \pmod{p^{2r}}$. First, $A_0 = \sum_{n<m} \sigma_r(n) K_0(n) \equiv 0 \pmod{p^{2r}}$ by Lemma 2.5. Let $0 \le s \le r-2$, $s' = r-s-1 \ge 1$, $L = L_{s+1}$ and $q = \lfloor n/p \rfloor$. Splitting the j-sum of $K_{s+1}(n)$ into scaled and unscaled j and using Lemmas 8.11 and 8.9,

$$K_{s+1}(n) = \sum_{j_1} \Psi_s(q, j_1)(1 - L\rho_s(n, j_1)) + L \sum'_j \frac{1}{j} \hat{\Xi}_{s+1}(n, j) + p^{2s+2}(\text{integral}).$$

Multiply by $\sigma_{s'}(n) \equiv 0 \pmod{p^{2s'}}$ and sum over $n < L_{s+1}$; the error is $\equiv 0 \pmod{p^{2s'+2s+2}} = \pmod{p^{2r}}$. Since $\sum_{n\in B_1(q)} \sigma_{s'}(n) = \sigma_{s'+1}(q)$,

$$A_{s+1} \equiv A_s - LX_s + LY_s \pmod{p^{2r}},$$

$$X_s := \sum_n \sigma_{s'}(n) \sum_{j_1} \Psi_s(q, j_1)\rho_s(n, j_1), \qquad Y_s := \sum_n \sigma_{s'}(n) \sum'_j \frac{\hat{\Xi}_{s+1}(n, j)}{j}.$$

It remains to show $X_s \equiv Y_s \equiv 0 \pmod{p^{2r-s-1}}$, since $v_p(L) = s+1$.

- $X_s = X_{s,1} - X_{s,2} + 2X_{s,3}$ according to ρ_s. First, $X_{s,1} = \sum_q c_q K_s(q)$ with $c_q = \sum_{n\in B_1(q)} \sigma_{s'}(n) H'_n$, whose block sums are $\equiv 0 \pmod{p^{t+1+2s'}}$ by Lemma 8.4; Lemma 8.7 applied to $c_q/p^{2s'+1}$ gives $X_{s,1} \equiv 0 \pmod{p^{2s'+1+s}} = \pmod{p^{2r-s-1}}$. Next, $X_{s,2} = \sum_q \sigma_{s'+1}(q)\tilde{K}^{(1)}_s(q)$, and the weights $\sigma_{s'+1}(q)/p^{2s'+2}$ have block sums $\equiv 0 \pmod{p^{2t}}$, so $X_{s,2} \equiv 0 \pmod{p^{2s'+2+s}}$ by Lemmas 8.7 and 8.10(4). Finally, as for $\hat{X}_3$, $X_{s,3} = X_{s,2} + \sum_{t\ge0}(-p)^t \sum_n \sigma_{s'}(n) S_{t+1}(n) K^{[t]}_s(q)$; the term $t = 0$ is $X_{s,1}$, and for $t \ge 1$ the term is $\equiv 0 \pmod{p^{t+2s'+s}}$ by Lemmas 2.5 and 8.10(3).
- For Y_s apply Lemma 8.8 at level $s+1$ to the tower $\hat{\Xi}_{s+1}; \Xi^{(\cdot)}_s; \Xi^{(\cdot)}_{s-1}; \dots$ of Lemma 8.10(2), with the top weights $a^{(0)}_{n,j} = p^{-2s'}\sigma_{s'}(n)/j$ for $p \nmid j$ (0 otherwise, and $a^{(1)} = 0$). They are integral, and their cell sums are $\equiv 0 \pmod{p^t}$ by Lemma 8.6(Y). Hence $\sum a^{(0)}_{n,j} \hat{\Xi}_{s+1}(n, j) \equiv 0 \pmod{p^{s+1}}$, that is $Y_s \equiv 0 \pmod{p^{2s'+s+1}} = \pmod{p^{2r-s-1}}$.

By induction $A_{r-1} \equiv 0 \pmod{p^{2r}}$, and with (6) this proves $\Sigma_K \equiv 0 \pmod{p^{3r}}$.

8.7. **Class J.** Class J is class K with the roles of the two indices exchanged; we give the argument in full, since the kernels are not symmetric. For $p \nmid j$ and $k = pk_1$, the identity $\binom{N+j-1}{j} = (-1)^j \binom{-N-1}{j} \frac{N}{N+j}$ of Lemma 8.9 gives $\tilde{S}(N; pk_1, j) = \frac{N}{N+j} \Phi_r(pk_1, j)$, where

$$\Phi_u(k, j) := \binom{L_u - 1}{k}^2 (-1)^k \binom{-L_u}{k} \binom{L_u - 1}{k+j}^2 (-1)^j \binom{-L_u - 1}{j}, \qquad K'_u(j) := \sum_{k\ge0} \Phi_u(k, j).$$

Put $\hat{K}'_r(j) := \sum_{k_1} \Phi_r(pk_1, j)$. The following properties are proved exactly as Lemmas 8.9, 8.10 and 8.11, by Lemma 2.6 applied factor by factor.

(J1) For $p\nmid k$, $\Phi_u(k,j)=\frac{L_u}{k}\hat\Xi'_u(k,j)$ with

$$\hat\Xi'_u(k,j):=\binom{L_u-1}{k}^2(-1)^{k-1}\binom{-L_u-1}{k-1}\binom{L_u-1}{k+j}^2(-1)^j\binom{-L_u-1}{j},$$

and $\hat\Xi'_u(k,j)\equiv\Xi^{(\delta)}_{u-1}(\lfloor k/p\rfloor,\lfloor j/p\rfloor)\pmod{p^u}$ with $\delta=[(k \bmod p)+(j \bmod p)\ge p]$. Indeed the four factors reduce to $\binom{L'-1}{\lfloor k/p\rfloor}^2$, $(-1)^{\lfloor k/p\rfloor}\binom{-L'-1}{\lfloor k/p\rfloor}$, $\binom{L'-1}{\lfloor k/p\rfloor+\lfloor j/p\rfloor+\delta}^2$ and $(-1)^{\lfloor j/p\rfloor}\binom{-L'-1}{\lfloor j/p\rfloor}$, which are the factors of $\Xi^{(\delta)}_{u-1}$ with the arguments in this order.

(J2) $\Phi_u(pk_1,j)\equiv\Phi_{u-1}(k_1,\lfloor j/p\rfloor)\pmod{p^u}$. Hence $\hat K'_r(j)\equiv K'_{r-1}(\lfloor j/p\rfloor)\pmod{p^r}$, and (K'_u) is a tower, the unscaled k vanishing modulo p^u by (J1).

(J3) The moments $K'^{[t]}_u(j):=\sum_k k^t\Phi_u(k,j)$ are $\equiv0\pmod{p^u}$ for $t\ge1$, and $\tilde K'^{(d)}_u(j):=\sum_k\Phi_u(k,j)H'_{p^dk}$ satisfies $\tilde K'^{(d)}_u(j)\equiv\tilde K'^{(d+1)}_{u-1}(\lfloor j/p\rfloor)\pmod{p^u}$ for $d\ge1$.

(J4) With $L=L_{u+1}$, exactly $\Phi_{u+1}(pk_1,j)=\Phi_u(k_1,\lfloor j/p\rfloor)\,\bar P_{pk_1}P^2_{pk_1}P^2_{pk_1+j}\bar P_j$, so

$$\Phi_{u+1}(pk_1,j)\equiv\Phi_u(k_1,\lfloor j/p\rfloor)(1-L\rho'_u(j,k_1))\pmod{p^{2u+2}}$$

with $\rho'_u(j,k_1):=-H'_j+H'_{pk_1}+2H'_{j+pk_1}$.

As in (6), $\Sigma_J\equiv N{\sum}'_{j<N}\hat K'_r(j)/j-N^2{\sum}'_{j<N}\hat K'_r(j)/j^2\pmod{p^{3r}}$. The second sum is $\equiv0\pmod{p^r}$ by (J2) and Lemmas 2.4 and 8.7. For the first, put $A'_s:=\sum_{j<L_s}\sigma_{r-s}(j)K'_s(j)$; by (J4),

$$\sum'_{j<N}\frac{\hat K'_r(j)}{j}\equiv A'_{r-1}-N\hat X'\pmod{p^{2r}},\qquad \hat X':=\sum'_{j<N}\frac1j\sum_{k_1}\Phi_{r-1}(k_1,\lfloor j/p\rfloor)\rho'_{r-1}(j,k_1).$$

Split $\hat X'=-\hat X'_1+\hat X'_2+2\hat X'_3$ according to ρ'. Then $\hat X'_1=\sum_{j'}c_{j'}K'_{r-1}(j')$ with $c_{j'}={\sum}'_{j\in B_1(j')}H'_j/j$, which is $\equiv0\pmod{p^r}$ by Lemmas 8.5 and 8.7; $\hat X'_2=\sum_{j'}\sigma_1(j')\tilde K'^{(1)}_{r-1}(j')\equiv0\pmod{p^{r+1}}$ by Lemmas 2.5 and 8.7 and (J3); and $H'_{j+pk_1}=H'_{pk_1}+\sum_{t\ge0}(-pk_1)^tS_{t+1}(j)$ gives $\hat X'_3=\hat X'_2+\hat X'_1+\sum_{t\ge1}(-p)^t{\sum}'_j\frac1jS_{t+1}(j)K'^{[t]}_{r-1}(\lfloor j/p\rfloor)\equiv0\pmod{p^r}$ by (J3). The signs in ρ' play no role.

For A'_{r-1}: $A'_0\equiv0\pmod{p^{2r}}$ by Lemma 2.5, and for $0\le s\le r-2$, with $s'=r-s-1$, $L=L_{s+1}$ and $q=\lfloor j/p\rfloor$, (J4) and (J1) give

$$A'_{s+1}\equiv A'_s-LX'_s+LY'_s\pmod{p^{2r}},$$

$$X'_s:=\sum_j\sigma_{s'}(j)\sum_{k_1}\Phi_s(k_1,q)\rho'_s(j,k_1),\qquad Y'_s:=\sum_j\sigma_{s'}(j){\sum}'_k\frac{\hat\Xi'_{s+1}(k,j)}{k}.$$

$X'_s\equiv0\pmod{p^{2r-s-1}}$ follows piece by piece exactly as for X_s, using Lemmas 8.4, 2.5, 8.7 and (J3). For Y'_s apply Lemma 8.8 at level $s+1$ with $(q,q')=(k,j)$, to the tower $\hat\Xi'_{s+1};\Xi^{(\cdot)}_s;\dots$ of (J1) and Lemma 8.10(2), with top weights $p^{-2s'}\sigma_{s'}(j)/k$ for $p\nmid k$. Their cell sums are the sums of Lemma 8.6(Y) with the two indices exchanged, hence $\equiv0\pmod{p^t}$; so $Y'_s\equiv0\pmod{p^{2s'+s+1}}$. This proves $\Sigma_J\equiv0\pmod{p^{3r}}$.

8.8. **Class B.** For $p\nmid k$ and $p\nmid j$, Lemma 8.9 twice gives

$$\tilde S(N;k,j)=\frac{N}{N+k}\cdot\frac{N}{j}\hat\Xi_r(k,j)=\frac{N^2}{kj}\hat\Xi_r(k,j)-\frac{N^3}{k^2j}\Big(1+\frac Nk\Big)^{-1}\hat\Xi_r(k,j),$$

so $\Sigma_B \equiv N^2 {\sum_{k,j}}' \hat{\Xi}_r(k,j)/(kj) \pmod{p^{3r}}$. Lemma 8.8 at level r, with the tower $\hat{\Xi}_r; \Xi_{r-1}^{(\cdot)}; \dots$ and the weights $1/(kj)$ for $p \nmid kj$, whose cell sums vanish modulo p^t by Lemma 8.6(B), gives $\Sigma_B \equiv 0 \pmod{p^{3r}}$.

8.9. **Class S.** For $k = pk'$, $j = pj'$, Lemma 2.6 (with m and $-m$) gives exactly $\tilde{S}(N; pk', pj') = \tilde{S}(M; k', j')E$ with $E = P_{pk'}^2 \bar{P}_{pk'} P_{p(k'+j')}^2 \bar{P}_{pj'}$ for $L = N$; the signs cancel since $pk' - k'$ is even. Hence $\Sigma_S - A(M-1) = \sum_{k',j'} \tilde{S}(M; k', j')(E-1)$. Let $a = v_p(k')$ and $b = v_p(j')$ (with $v_p(0) = \infty$).

Step 1: second and higher order. By Lemma 8.1, $E - 1 = -N\rho_S + N^2 e_2 + \sum_{d \ge 3} N^d e_d$ with integral e_d and $\rho_S = H'_{pk'} + 2H'_{p(k'+j')} - H'_{pj'}$. For $x \in \{k', k'+j', j'\}$ with $x \ge 1$ we have $v_p(px) \ge 1 + \min(a,b)$, so $S_2(px) \equiv 0 \pmod{p^{1+\min(a,b)}}$ and $\rho_S \equiv 0 \pmod{p^{2+2\min(a,b)}}$ (Lemmas 2.4 and 2.5); hence $v_p(N^2 e_2) \ge 2r + 1 + \min(a,b)$. On the other hand $\binom{M+k'-1}{k'} = \frac{M}{k'}\binom{M+k'-1}{k'-1}$ is divisible by $p^{(r-1-a)^+}$, and likewise for j', so $v_p(\tilde{S}(M; k', j')) \ge (r-1-a)^+ + (r-1-b)^+ \ge (r - 1 - \min(a,b))^+$. Thus the terms of order ≥ 2 are $\equiv 0 \pmod{p^{3r}}$ (if $k' = j' = 0$ then $E = 1$), and

$$\Sigma_S - A(M-1) \equiv -NT_S \pmod{p^{3r}}, \qquad T_S := \sum_{k',j'} \tilde{S}(M; k', j')\rho_S(k', j').$$

It remains to show $T_S \equiv 0 \pmod{p^{2r}}$.

Step 2: decoupling. Exactly, $H'_{p(k'+j')} - H'_{pk'} = {\sum_{\mu \le pj'}}' \frac{1}{pk'+\mu} = H'_{pj'} - pk'Z$ with $Z := {\sum_{\mu \le pj'}}' \frac{1}{\mu(\mu+pk')}$, so $\rho_S = 3H'_{pk'} + H'_{pj'} - 2pk'Z$. We have $v_p(Z) \ge 1 + b$: indeed $Z = \sum_{t \ge 0} (-pk')^t S_{t+2}(pj')$, and $S_{t+2}(pj') \equiv 0 \pmod{p^{1+b-e(t+2)}}$ by Lemma 2.4, since pj' consists of full blocks of length p^{1+b}. The t-th term thus has valuation $\ge 1+b+t-e(t+2) \ge 1+b$, because $e(t+2) = 0$ unless $(p-1) \mid t+2$, which forces $t+2 \ge 4$ and then $e(t+2) = 1 + v_p(t+2) \le t$. Consequently $v_p(N \cdot pk'Z \cdot \tilde{S}(M; k', j')) \ge r + (1+a) + (1+b) + (r-1-a)^+ + (r-1-b)^+ \ge 3r$, and

$$T_S \equiv 3T_1 + T_2 \pmod{p^{2r}}, \qquad T_1 := \sum_{k',j'} \tilde{S}(M; k', j')H'_{pk'}, \quad T_2 := \sum_{k',j'} \tilde{S}(M; k', j')H'_{pj'}.$$

Step 3: $T_1 \equiv 0 \pmod{p^{2r}}$. Write $T_1 = \sum_{k' \ge 1} H'_{pk'} \binom{M-1}{k'}^2 \binom{M+k'-1}{k'} G_1(k')$ with $G_1(k') := \sum_{j'} \binom{M-1}{k'+j'}^2 \binom{M+j'-1}{j'}$, and stratify by $\sigma := v_p(pk') \ge 1$, $k' = p^{\sigma-1}\kappa$ with $p \nmid \kappa$. If $\sigma \ge r$, then $H'_{pk'} \equiv 0 \pmod{p^{2r}}$ by Lemma 2.5. Let $1 \le \sigma \le r-1$. Then $\binom{M+k'-1}{k'} = \frac{mp^{r-\sigma}}{\kappa}(-1)^{k'-1}\binom{-M-1}{k'-1}$. Reducing the remaining factors $\sigma - 1$ times by Lemma 2.6, modulo $p^{r-\sigma+1}$,

$$\binom{M-1}{k'}^2 \equiv \binom{mp^{r-\sigma}-1}{\kappa}^2, \qquad (-1)^{k'-1}\binom{-M-1}{k'-1} \equiv (-1)^{\kappa-1}\binom{-mp^{r-\sigma}-1}{\kappa-1},$$
$$G_1(k') \equiv G_1^{(r-\sigma)}(\kappa),$$

where $G_1^{(u)}(x) := \sum_j \binom{mp^u-1}{x+j}^2 \binom{mp^u+j-1}{j}$. The last congruence holds because $(G_1^{(u)})_u$ is a tower: for $p \nmid j$, $\binom{mp^u+j-1}{j} = \frac{mp^u}{j}\binom{mp^u+j-1}{j-1} \equiv 0 \pmod{p^u}$, and for $j = pj_1$, Lemma 2.6 gives $\binom{mp^u-1}{x+pj_1}^2 \equiv \binom{mp^{u-1}-1}{\lfloor x/p \rfloor + j_1}^2$ and $\binom{mp^u+pj_1-1}{pj_1} \equiv \binom{mp^{u-1}+j_1-1}{j_1} \pmod{p^u}$. The reduction errors are multiples of $p^{r-\sigma+1}$ and meet the explicit factor $p^{r-\sigma}$ and $H'_{pk'} \equiv 0 \pmod{p^{2\sigma}}$, so they are

$\equiv 0 \pmod{p^{2r}}$. The stratum is therefore

$$\equiv mp^{r-\sigma} \sum_{\kappa<mp^{r-\sigma}}{}' \frac{H'_{p^\sigma\kappa}}{\kappa} R^{(r-\sigma)}(\kappa) \pmod{p^{2r}},$$

where

$$R^{(u)}(\kappa) := \binom{mp^u-1}{\kappa}^2 (-1)^{\kappa-1} \binom{-mp^u-1}{\kappa-1} G_1^{(u)}(\kappa).$$

The weights $a_\kappa := p^{-2\sigma} H'_{p^\sigma\kappa}/\kappa$ ($p \nmid \kappa$; 0 otherwise) are integral by Lemma 2.5, and their block sums are $\equiv 0 \pmod{p^t}$ by Lemma 8.5 with $s = \sigma$. For $p \nmid \kappa$, Lemma 2.6 gives $R^{(u)}(\kappa) \equiv \hat{R}^{(u-1)}(\lfloor \kappa/p \rfloor) \pmod{p^u}$ with

$$\hat{R}^{(w)}(q) := \binom{mp^w-1}{q}^2 (-1)^q \binom{-mp^w-1}{q} G_1^{(w)}(q),$$

and $(\hat{R}^{(w)})_w$ is a tower by Lemma 2.6 and the tower property of G_1. Lemma 8.7 (with the top function needed only on units) gives $\sum_\kappa{}' a_\kappa R^{(r-\sigma)}(\kappa) \equiv 0 \pmod{p^{r-\sigma}}$, so the stratum is $\equiv 0 \pmod{p^{(r-\sigma)+2\sigma+(r-\sigma)}} = \pmod{p^{2r}}$.

Step 4: $T_2 \equiv 0 \pmod{p^{2r}}$. Write $T_2 = \sum_{j'\geq 1} H'_{pj'} \binom{M+j'-1}{j'} G_2(j')$ with

$$G_2(j') := \sum_{k'} \binom{M-1}{k'}^2 \binom{M+k'-1}{k'} \binom{M-1}{k'+j'}^2,$$

and stratify by $\sigma = v_p(pj')$, $j' = p^{\sigma-1}\iota$. The strata with $\sigma \geq r$ vanish as before. For $\sigma \leq r-1$, $\binom{M+j'-1}{j'} = \frac{mp^{r-\sigma}}{\iota}(-1)^{j'-1}\binom{-M-1}{j'-1}$, the last factor reduces to $(-1)^{\iota-1}\binom{-mp^{r-\sigma}-1}{\iota-1}$, and $G_2(j') \equiv G_2^{(r-\sigma)}(\iota) \pmod{p^{r-\sigma}}$ with $G_2^{(u)}(y) := \sum_k \binom{mp^u-1}{k}^2 \binom{mp^u+k-1}{k} \binom{mp^u-1}{k+y}^2$, a tower in y (the unscaled k vanish modulo p^u through $\binom{mp^u+k-1}{k}$, and the scaled $k = pk_1$ reduce by Lemma 2.6 with $\lfloor (pk_1+y)/p \rfloor = k_1 + \lfloor y/p \rfloor$). The rest is as in Step 3.

Hence $T_S \equiv 0 \pmod{p^{2r}}$ and $\Sigma_S \equiv A(M-1) \pmod{p^{3r}}$.

8.10. **Conclusion.** $A(N-1) - A(M-1) = (\Sigma_S - A(M-1)) + \Sigma_K + \Sigma_J + \Sigma_B \equiv 0 \pmod{p^{3r}}$. For $r = 1$ the inductions over s are empty (only $A_0 \equiv A'_0 \equiv 0 \pmod{p^2}$ and $\hat{X} \equiv \hat{X}' \equiv 0 \pmod{p}$ are used), every stratum of Steps 3 and 4 has $\sigma \geq r$, and Lemma 8.8 is used at level 1. Nothing used $p \nmid m$. □

Remark 8.12. Lemma 8.5 is needed in its strong form: modulo p^{t+s} only, Step 3 would give $p^{2r-\sigma}$.

9. On the printed proof of Beukers (1985)

We compared [3] with the publisher's scan (pp. 148–154).

(1) On p. 149 the proof of Lemma 4(i) reduces it to $\sum_{n=0}^{p^t-1} n^k \sum_{\lambda\leq n}{}' \frac{1}{\lambda} \equiv 0 \pmod{p^t}$ for all $k \geq 0$. This congruence is false: for $k = 0$ the sum equals $p^t H'_{p^t-1} - (p^t - p^{t-1}) \equiv p^{t-1} \pmod{p^t}$ (for instance $\frac{77}{12} \equiv 1 \pmod 5$), and for $k = 1, 2$ it has valuation $t-1$ in all cases we computed. The Abel summation that follows drops the condition $p \nmid n+1$. Lemma 4(i) itself is true (Lemma 8.4).

(2) Lemma 4(ii) is stated modulo p^{t+s} and proved for the block $i = 0$ only. On p. 154 it is applied with the weights $p^{-s}k^{-1}\sum'_{\lambda\le kp^s}\lambda^{-1}$; this gives the term (13) modulo p^{2r-s}, whereas p^{2r} is required. With the normalisation p^{-2s} and Lemma 8.5 the count is $(r-s)+2s+(r-s)=2r$.

Beukers' theorem is unaffected (it also follows from [8, 23]).

10. Verification

The Dixon-type sum and the factorial ratios. *Numerics.* All statements were tested in exact arithmetic directly from the OEIS definitions; the exponents are attained (Theorem 1.1: $v_p(a(p^k)-a(p^{k-1}))=3k$ for $p\in\{5,11,13,\dots,43\}$ and $3k+1$ for $p=7$; Theorem 1.3: slack 0 or 1 for $p\in\{5,7,11,13\}$). The identity (2) was checked exactly for $p\le 13$, odd $n\le 11$, $k\le 2$.

Lean. The Lean 4/Mathlib project accompanying this paper [12] proves the Formal Conjectures statements of A361711 (Theorem 1.1 for $n=1$), A364175 and A364178 (Theorem 1.3, including the integrality of A364178, which in the benchmark is defined through the real Γ function and rounding). Each benchmark file was copied verbatim into a separate check file whose only `sorry` is replaced by our theorem; all compile, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The extension to odd $n>1$ and the integrality of A364172 and A364179 are not yet formalised.

The Apéry-like sequences. *Numerics.* Both theorems were tested in exact arithmetic from the OEIS definitions. For A376462, both congruences were checked up to $r=2$ for $p\le 31$, $r=3$ for $p\le 13$, $r=4$ for $p=7$ and $r=5$ for $p=5$ ($N=3125$); the exponent $3r$ is attained. Every lemma of Section 8 was also checked separately, and the bounds are attained (slack 0) at every stage.

Lean. The Lean 4/Mathlib project [12] proves the Formal Conjectures statements of A363983 and A376462 (the latter with both congruences). Each benchmark file was copied verbatim into a check file whose only `sorry` is replaced by our theorem; it compiles, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The A363983 definition uses `Int.toNat`, and the formal proof establishes positivity. The formalisation uses a Kazandzidis lemma from an accepted proof of the OEIS Open evaluation of Epoch AI [1, 10], included with attribution.

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Declarations

Use of AI tools. We used a large language model to help draft the text and proofs. The authors checked all of the work and are responsible for it.

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Competing interests. The authors have no competing interests to declare.

Data availability. Scripts, outputs and Lean files are available at [12].

Appendix A. The certificate polynomial

$$\begin{aligned}P(n,k) = {} & 84n^8-234kn^7+520n^7+261k^2n^6-1215kn^6+1306n^6-135k^3n^5+1101k^2n^5-2590kn^5 \\ & +1772n^5+27k^4n^4-441k^3n^4+1934k^2n^4-3042kn^4+1454n^4+63k^4n^3-606k^3n^3+1870k^2n^3 \\ & -2183kn^3+752n^3+60k^4n^2-434k^3n^2+1062k^2n^2-1000kn^2+268n^2+26k^4n-160k^3n \\ & +344k^2n-296kn+80n+4k^4-24k^3+52k^2-48k+16.\end{aligned}$$

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